NPTEL ONLINE CERTIFICATION COURSES



Course Name: Software Project Management

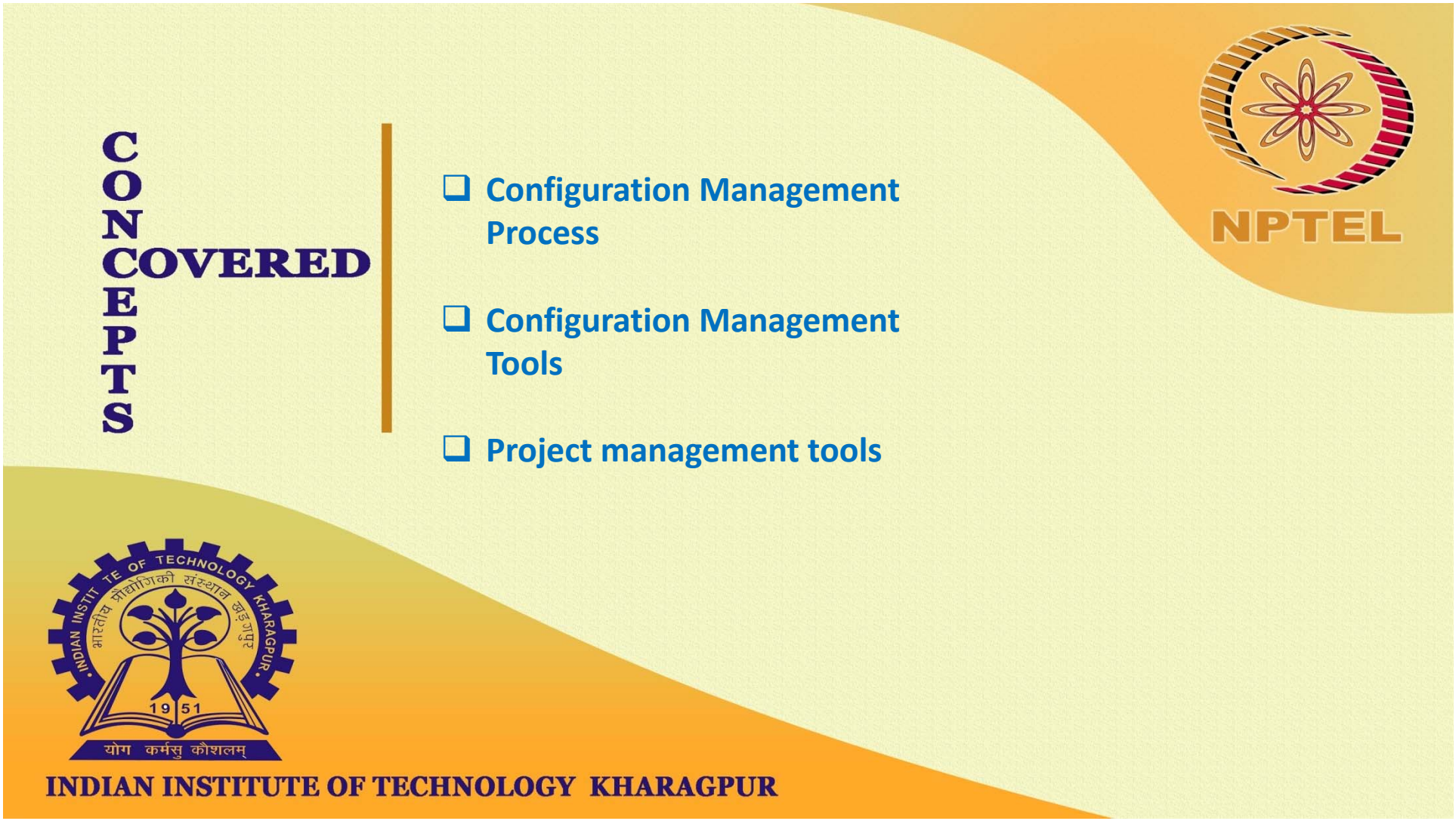
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Department : Computer Sc. & Engg. NIT Rourkela

TOPIC: Project Monitoring and control



INDIAN INSTITUTE OF TECHNOLOGY KHARAGPUR


CONCEPTS COVERED
Configuration Management Process
Configuration Management Tools
Project management tools
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Configuration management process

- Configuration management is carried out through the following two principal activities.
  - Configuration identification
    - Involves deciding which parts of the system should be kept under configuration management
  - Configuration control
    - Used to ensure that changes to a system occur smoothly.

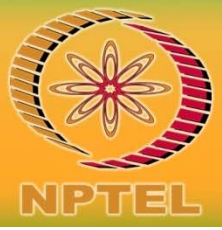




Configuration identification

- Project managers normally classify the work products associated with a software development process into three main categories
  - Controlled
  - Pre-controlled
  - Uncontrolled
- Controlled work products are those that are put under configuration control
  - Team members must follow some formal procedures to change these.
- Pre-controlled work products are not yet under configuration control, but will eventually be under configuration control.





Configuration identification cont …

- Uncontrolled work products will not be subject to configuration control.
- Controllable work products include both controlled and pre-controlled work products.





Configuration identification cont …

- Typical controllable work products include the following
  - Requirements specification document
  - Design documents
  - Tools used to build the system such as compilers, linkers, lexical analysers, parsers, etc.
  - Source code for each module
  - Test cases
  - Problem reports





Configuration control

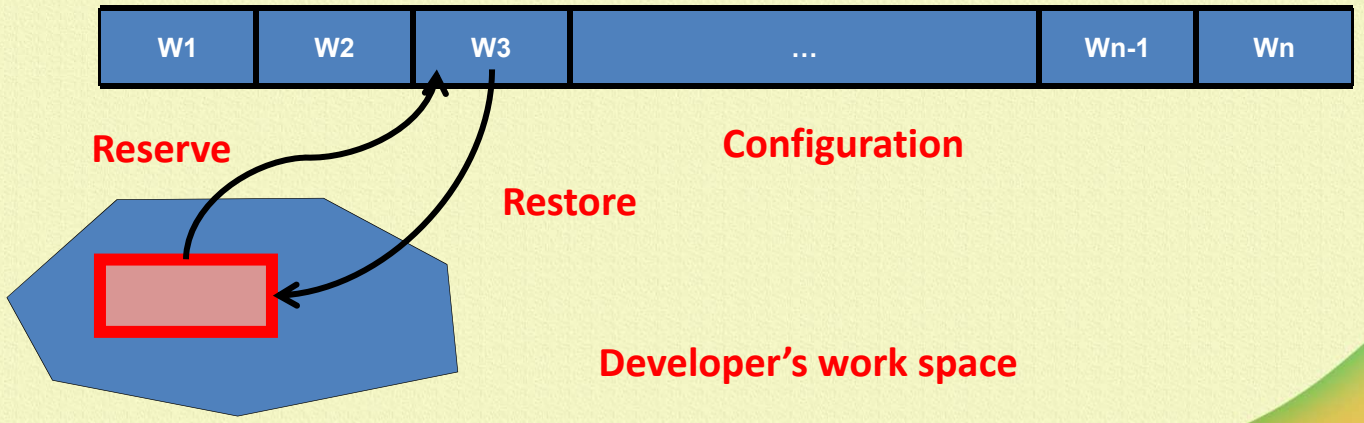
- Configuration control is the part of configuration management system that most directly affects the day-to-day operations of developers.
- Configuration control allows only authorized changes to the controlled objects and prevents unauthorized changes.
- The project manager can give permission to some members to be able to change or access specific work products.
- In order to change a controlled work product such as a code module, a developer can get a private copy of the module through a reserve operation.





Configuration Control cont …

- Two main operations:
  - Reserve
  - Restore



Configuration Control cont …

- Configuration management tools allow only one team member to reserve a module at any time.
- Once a work product is reserved, it does not allow anyone else to reserve this module until the reserved module is restored.
- Thus, by preventing more than one developer to simultaneously reserve a module, the problems associated with concurrent access are taken care of.





Modifications to a work product under configuration control

- When developers need to change a work product, they first make a reserve request.
- A reserve request by a team member is honoured only if appropriate authorization has been given by the project manager to that member for the specific work product.
- After the reserve command successfully executes, a private copy of the work product is created in their local directory.
- Then they carry out all necessary changes to the work product on their private copy.





Modifications to a work product under configuration control cont...

- Once they have satisfactorily completed all necessary changes, the changes need to be restored in configuration management repository.
- However, restoring the changed work product to the system configuration requires the permission of a change control board (CCB).
- CCB is usually constituted from among the development team members.
- For every change that needs to be carried out, the CCB reviews the changes made to the controlled work product and certifies certain aspects about the change such as:

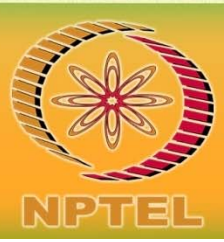




Modifications to a work product under configuration control cont...

- Change is well-motivated
- Developer has considered and documented the effects of the change
- Changes interact well with the changes made by other developers
- Appropriate people (may be CCB) have validated the change, e.g. someone has tested the changed code, and has verified that the change is consistent with the need.





Modifications to a work product under configuration control cont...

- The change control board (CCB) is seldom a group of people.
- Except for vary large projects, the functions of the change control board are normally discharged solely by the project manager or some senior member of the development team.
- Once the CCB reviews the changes to the module and approves them, the project manager updates the old configuration item through a restore operation.





Modifications to a work product under configuration control cont...

- A configuration control tool does not allow a developer to replace a work product in the configuration with his local copy unless he gets an authorization form the CCB.
- Therefore, incompletely modified or improperly modified work products cannot be updated in the configuration.





Release Management

- Release Management process systemizes the work carried out by developers to provide a new release of a software and on the part of the users to smoothly obtain and use a new release.
- The release process should involve minimal effort on the part of the developer to upload a new release of a software and on the part of the users to effortlessly download and install it.





Release Management cont ...

- For example, when a new version of the system is to be released, a tool should automatically determine the changed components, the dependencies, and all interdependent components should be retrievable as a group so that there is no possibility of inconsistency.
- Also retrievals of unnecessary and unchanged components should be avoided.





Open source configuration management tools

- Two popular configuration management tools on UNIX systems
  - SCCS (Source Code Control System)
  - RCS (Revision Control System)
- SCCS or RCS can be used for controlling and managing different versions of text files.
- SCCS and RCS do not handle binary files (i.e., executable files, documents, files containing diagrams, etc.).
- SCCS and RCS provide an efficient way of storing versions that minimize the amount of occupied disk space.





Open source configuration management tools cont...

- Suppose, a module MOD is present in three versions MOD1.1, MOD1.2, and MOD1.3.
- Then SCCS and RCS stores the original module MOD1.1 together with changes needed to transform MOD1.1 into MOD1.2 to MOD1.3.
- The changes needed to transform each baseline file to the next version are stored and are called deltas.
- The main reason behind storing the deltas rather than storing the full revision files is to save disk space.

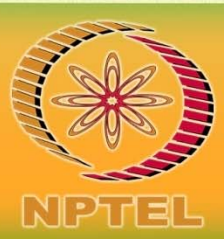




Open source configuration management tools cont...

- The change control facilities provided by SCCS and RCS include the ability to incorporate restrictions on the set of individuals who can create new versions, and facilities for checking components in and out (i.e., reserve and restore operations).
- Individual developers check out components and modify them.
- After they have made all the necessary changes to a component, and after these changes have been reviewed, they check in the changed module into SCCS or RCS.





SCCS (Source Code Control System)

- SCCS is a configuration management tool:
  - available on Unix systems:
  - helps control and manage text files.
  - also implements an efficient way of storing versions:
    - minimizes the amount of occupied disk space.





SCCS cont …

- Suppose a module is present in 3 versions:
  - MOD1.1, MOD1.2, and MOD1.3.
- SCCS stores the original module MOD1.1
  - together with changes needed to transform it into MOD1.2 and MOD1.3.
  - the modifications are called deltas.





SCCS Features

- Access control facilities provided by SCCS include:
  - facilities for checking components in and out.
  - Individual developers check out components and modify them.
    - after they have changed a module as required and the module has been successfully tested,
    - they check in the changed module into SCCS.





SCCS Features

- Revisions are denoted by numbers in ascending order,
  - e.g, 1.1, 1.2, 1.3 etc.
- It is also possible to create variants of a component
  - by creating a fork in the development history.

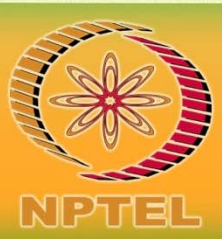




Some project management tools

- Ganttproject is freeware (GPL-Licensed) project management software that runs under the Windows, Linux and Mac operating systems.
- Microsoft Project is the basic project management software from Microsoft Corporation. Advanced capabilities are supported through MS-Office Project Server, and MS-Office Project Web Access software.

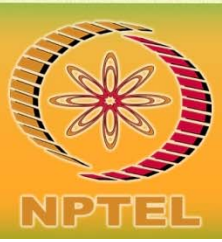




project management tools cont…

- Primavera project management software is a widely used suite of project management software. SureTrack is the entry-level software and Primavera 6 is the advanced software.
- Using SureTrack's Project KickStart a project manager can define project phases, establish goals, anticipate obstacles, and delegate assignments.

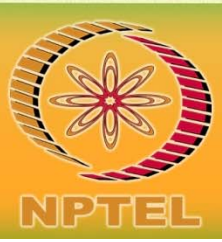




Features of the project management tools

Software	Portfolio management	Web-based	Scheduling	Cost management	Resource management	Open source
Ganttproject			√		√	√
Microsoft Project	√	√	√	√	√	
Primavera SureTrack	√	√	√	√	√	





Summary

- Explained briefly the configuration management process.
- Presented some configuration management tools such as SCCS and RCS.
- Also presented some project management tools.





References :

1. B. Hughes, M. Cotterell, R. Mall, *Software Project Management*, Sixth Edition, McGraw Hill Education (India) Pvt. Ltd., 2018.
2. R. Mall, *Fundamentals of Software Engineering*, Fifth Edition, PHI Learning Pvt. Ltd., 2018.






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